

PHASE 4: PROJECT PLANNING PHASE

Project Details

Particular	Details
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Project Planning Phase
Date	15 February 2025
Team ID	_____
Prepared By	Koppana Navya

4.1 Product Backlog & Sprint Planning

Introduction

Agile Scrum methodology was adopted to organize the project into manageable development iterations called **Sprints**. The project was divided into Epics, User Stories, and Story Points to estimate the effort required for each task.

Story Point Scale

Story Points Complexity

1	Very Easy
2	Easy
3	Moderate
5	Complex

Sprint 1

Epic 1 – Data Collection & Extraction

User Story No.	User Story / Task	Story Points
USN-1	Download Food Ordering Behaviour Dataset	2
USN-2	Load Dataset into Tableau Desktop	1

Epic 2 – Data Preparation

User Story No.	User Story / Task	Story Points
USN-3	Handle Missing Values	3
USN-4	Create Calculated Fields (Age Groups)	3
USN-5	Validate and Standardize Dataset	3

Sprint 1 Summary

Total Story Points = 2 + 1 + 3 + 3 + 3 = 12

Sprint 2

Epic 3 – Data Visualization

User Story No.	User Story / Task	Story Points
USN-6	Create KPI Cards	2
USN-7	Create Heat Map & Pie Chart	2
USN-8	Create Bar Charts & Donut Chart	3
USN-9	Create Revenue & Delivery Time Charts	3

Epic 4 – Dashboard Development

User Story No.	User Story / Task	Story Points
USN-10	Design Interactive Dashboard	5

Epic 5 – Story Development

User Story No.	User Story / Task	Story Points
USN-11	Develop Tableau Story	5

Sprint 2 Summary

Total Story Points = 2 + 2 + 3 + 3 + 5 + 5 = 20

Sprint 3

Epic 6 – Performance Testing

User Story No.	User Story / Task	Story Points
USN-12	Validate 50,000 Records	2
USN-13	Test Filters and Calculated Fields	2
USN-14	Verify Dashboard Performance	2

Epic 7 – Web Integration

User Story No.	User Story / Task	Story Points
USN-15	Publish Dashboard to Tableau Public	3
USN-16	Integrate Dashboard with Flask	5

Sprint 3 Summary

Total Story Points = 2 + 2 + 2 + 3 + 5 = 14

4.2 Product Backlog

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Data Collection	USN-1	Download Dataset	2	High	Koppana Navya
Sprint-1	Data Collection	USN-2	Load Dataset into Tableau	1	High	Koppana Navya
Sprint-1	Data Preparation	USN-3	Handle Missing Values	3	High	Koppana Navya
Sprint-1	Data Preparation	USN-4	Create Age Groups	3	High	Koppana Navya
Sprint-1	Data Preparation	USN-5	Validate Dataset	3	High	Koppana Navya
Sprint-2	Visualization	USN-6	KPI Cards	2	High	Koppana Navya

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-2	Visualization	USN-7	Heat Map & Pie Chart	2	Medium	Koppana Navya
Sprint-2	Visualization	USN-8	Bar & Donut Charts	3	High	Koppana Navya
Sprint-2	Visualization	USN-9	Revenue Charts	3	High	Koppana Navya
Sprint-2	Dashboard	USN-10	Interactive Dashboard	5	High	Koppana Navya
Sprint-2	Story	USN-11	Tableau Story	5	High	Koppana Navya
Sprint-3	Testing	USN-12	Data Validation	2	High	Koppana Navya
Sprint-3	Testing	USN-13	Filter Testing	2	High	Koppana Navya
Sprint-3	Testing	USN-14	Dashboard Testing	2	High	Koppana Navya
Sprint-3	Web Integration	USN-15	Tableau Public Publishing	3	Medium	Koppana Navya
Sprint-3	Web Integration	USN-16	Flask Integration	5	High	Koppana Navya

4.3 Project Tracker, Velocity & Burndown

Sprint Schedule

Sprint	Total Story Points	Duration	Sprint Start	Sprint End (Planned)	Story Points Completed	Release Date
Sprint-1	12	5 Days	15 Feb 2025	19 Feb 2025	12	19 Feb 2025

Sprint	Total Story Points	Duration	Sprint Start	Sprint End (Planned)	Story Points Completed	Release Date
Sprint-2	20	7 Days	20 Feb 2025	26 Feb 2025	20	26 Feb 2025
Sprint-3	14	5 Days	27 Feb 2025	03 Mar 2025	14	03 Mar 2025

Velocity Calculation

Total Story Points

Sprint-1 = **12**

Sprint-2 = **20**

Sprint-3 = **14**

Total Story Points = 46

Number of Sprints = 3

Velocity

Velocity = Total Story Points / Number of Sprints

Velocity = 46 / 3 = 15.33

Average Team Velocity ≈ 15 Story Points per Sprint

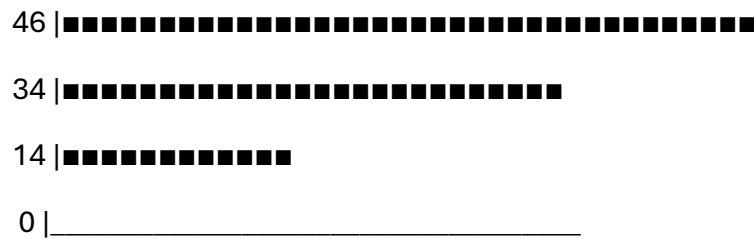
Burndown Analysis

The burndown chart tracks the remaining work against time for each sprint.

Sprint	Planned Story Points	Completed Story Points	Remaining Story Points
Sprint-1	12	12	34
Sprint-2	20	20	14
Sprint-3	14	14	0

Burndown Trend

Story Points



Sprint-1 Sprint-2 Sprint-3

The chart illustrates a steady reduction in pending work across all three sprints, indicating consistent progress and timely completion of planned tasks.

4.4 Sprint Review & Planning Summary

Sprint Outcomes

Sprint 1

- Dataset successfully collected from Kaggle.
- Dataset imported into Tableau.
- Data cleaning and preprocessing completed.
- Calculated Age Group field created.

Sprint 2

- Nine Tableau visualizations developed.
- Interactive dashboard created.
- Tableau Story prepared with multiple analytical scenes.

Sprint 3

- Performance testing completed.
- Dashboard validated using 50,000 records.
- Tableau workbook published to Tableau Public.
- Dashboard embedded into a Flask web application.

Project Planning Summary

The project was successfully executed using Agile Scrum principles. Dividing the work into three sprints enabled systematic completion of data collection, preprocessing,

visualization, testing, and deployment activities. Story point estimation and velocity tracking helped monitor progress and maintain an efficient development schedule. The project achieved all planned deliverables within the estimated sprint durations, resulting in a complete Tableau analytics solution with web integration.